

HR / Employee Analytics Summary Report

Perspective: Human Resources and Sales Performance.

Dataset: AdventureWorks 2022

1. Perspective & Focus Area

This analysis focuses on the **Human Resources and Employee Analytics** perspective at AdventureWorks Cycles. The primary goal is to understand the composition of the workforce headcount, demographics, compensation, and to also evaluate the sales team's performance by linking employee data to sales outcomes.

The analysis spans five key sub-areas: **workforce composition, demographic breakdown, compensation equity, sales representative performance, and organisational risk indicators.**

2. Tables Used & Rationale

Table	Purpose
HumanResources.Employee	Core employee attributes, tenure, leave
HumanResources.Department	Department names and group classification
HumanResources.EmployeeDeptHistory	Current dept assignment (EndDate IS NULL)
HumanResources.EmployeePayHistory	Pay rate history per employee
Sales.SalesPerson	YTD sales, quota, bonus, commission
Person.Person	First & last name lookup by BusinessEntityID
Sales.SalesOrderHeader	Order-level revenue by SalesPersonID

3. Key Assumptions

- Annual Pay derivation: PayFrequency = 2 means bi-weekly; annual pay = Rate × 26. PayFrequency = 1 means monthly; annual pay = Rate × 12. No other frequency codes were present in the data.
- Tenure calculation: Computed as (July 1, 2014 – HireDate) ÷ 365.25. This is treated as the analysis snapshot date since the latest ModifiedDate in the dataset is June 30, 2014.
- Current department: EmployeeDepartmentHistory records where EndDate IS NULL represent the employee's current (active) department assignment.
- All 290 employees have CurrentFlag = 1, meaning no terminated or inactive employees appear in the dataset — the workforce snapshot is entirely active.
- Three sales reps (Jiang, Alberts, Abbas) have no TerritoryID — assumed to be in non-field sales roles (e.g., inside sales or management). They are included in headcount but noted separately.
- Leave hours (VacationHours, SickLeaveHours) reflect balances as of the snapshot, not annual allocation. Values are used for relative comparison only.

Methodology, Insights & Recommendations

4. Methodology & Derivations

- Workforce Composition
- Headcount was counted by joining Employee → EmployeeDepartmentHistory (EndDate IS NULL) → Department. Gender and salaried/hourly splits were read directly from Employee.Gender and Employee.SalariedFlag (1=salaried, 0=hourly).
- Compensation Analysis
- The most recent pay record per employee was extracted by taking the maximum RateChangeDate from EmployeePayHistory. Annual pay was computed as Rate × 26 (bi-weekly) or Rate × 12 (monthly). Group and department averages were then calculated from these annualised figures.
- Sales Performance
- Total cumulative revenue per rep was computed by summing SubTotal in SalesOrderHeader grouped by SalesPersonID. YTD and LastYear figures came directly from Sales.SalesPerson. Reps were ranked by cumulative total revenue to identify consistent high performers beyond a single YTD snapshot.

5. Key Insights

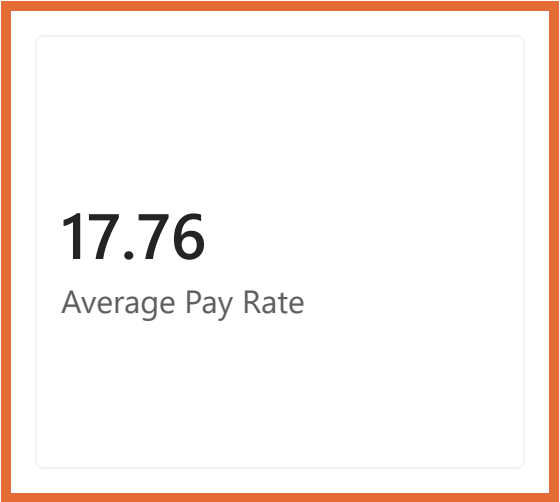
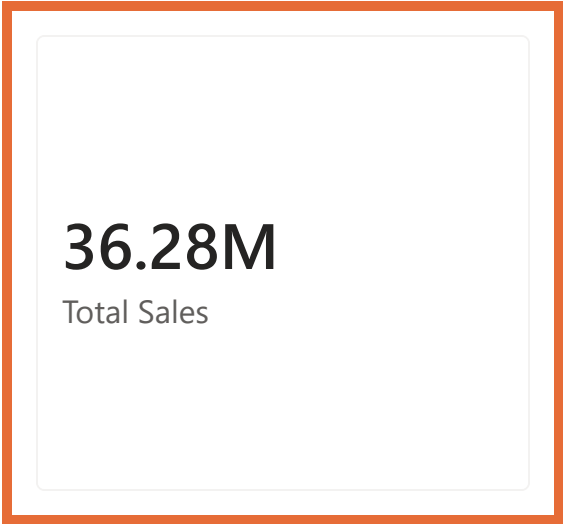
Finding	Metric	Risk/Opportunity
• Gender imbalance — 71% male workforce	• 84 female of 290 (29%)	• ⚠ High Risk
• 2009 hiring concentration — tenure cliff risk	• 148 hires in one year (51%)	• ⚠ High Risk
• Production dept underpaid vs. company avg	• \$217 avg vs. \$377 company avg	• ⚠ Moderate Risk
• Top 5 sales reps generate 55% of revenue	• \$10M+ each cumulative	• ✓ Strength
• R&D & Engineering highly compensated	• \$974–\$1,136 avg rate	• ✓ Strength
• 3 non-territory reps earning highest base pay	• \$1,250/bi-weekly each	• → Review
• Quality Assurance: 1 female vs. 10 male	• 9% female in QA dept	• ⚠ Diversity Gap

6. Recommendations

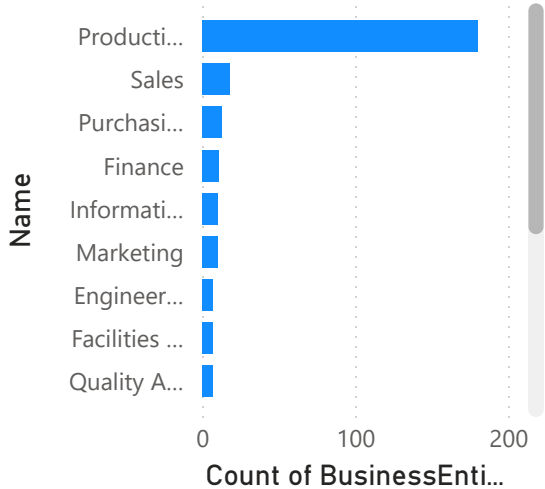
- Address gender diversity. A 29% female representation, falling to 9% in Quality Assurance and 25% in Manufacturing, suggests structural barriers in technical hiring. Set measurable diversity targets for technical departments and review job description language and recruitment channels.
- Protect and reward top sales talent. Linda Mitchell, Jillian Carson, and Michael Blythe collectively generate over \$29M in cumulative revenue. Structured retention packages (equity, career ladders, enhanced commission tiers) should be prioritised for top-5 performers.
- Expand R&D headcount. With only 14 employees in Research & Development the second-highest paid group there is a strong signal of innovation investment but limited scale. Targeted R&D hiring can strengthen the product pipeline without disproportionate cost impact.
- Review Production compensation. Production is the largest department (179 employees, 62% of workforce) but has the lowest average pay (\$217 bi-weekly). Benchmarking against industry standards and reviewing pay floors would improve retention and morale in the backbone of operations.

AdventureWorks HR & Employee Performance Analytics Dashboard

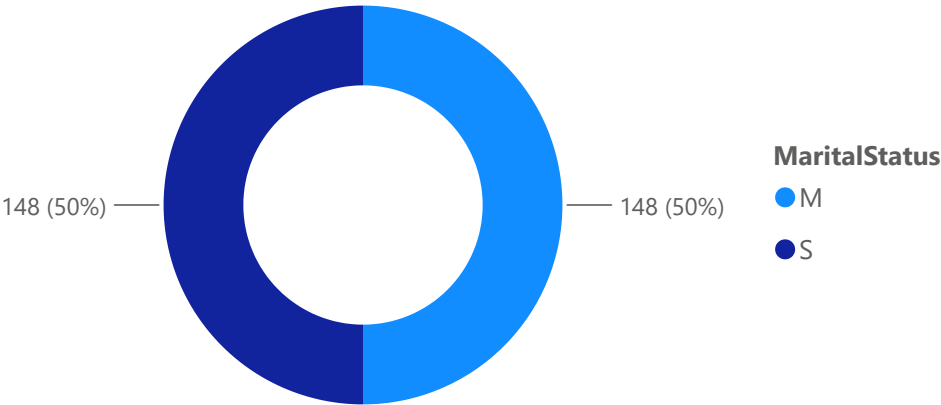
AdventureWorks HR & Employee Performance Analytics Dashboard



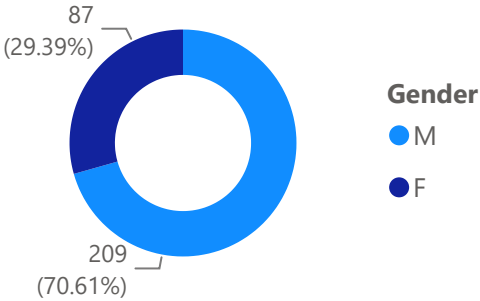
Employee Count by Department



Employee Distribution by Marital Status

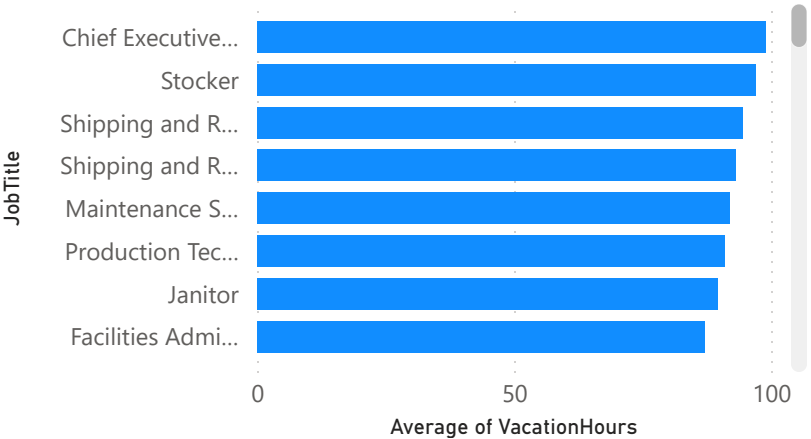


Employee Distribution by Gender

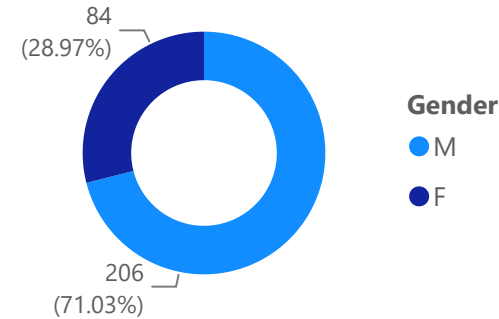




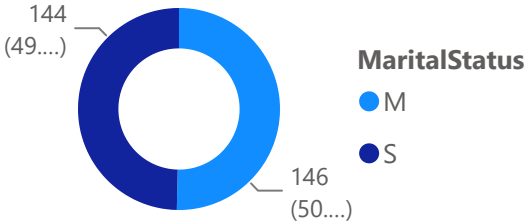
Average Vacation hours by JobTitle



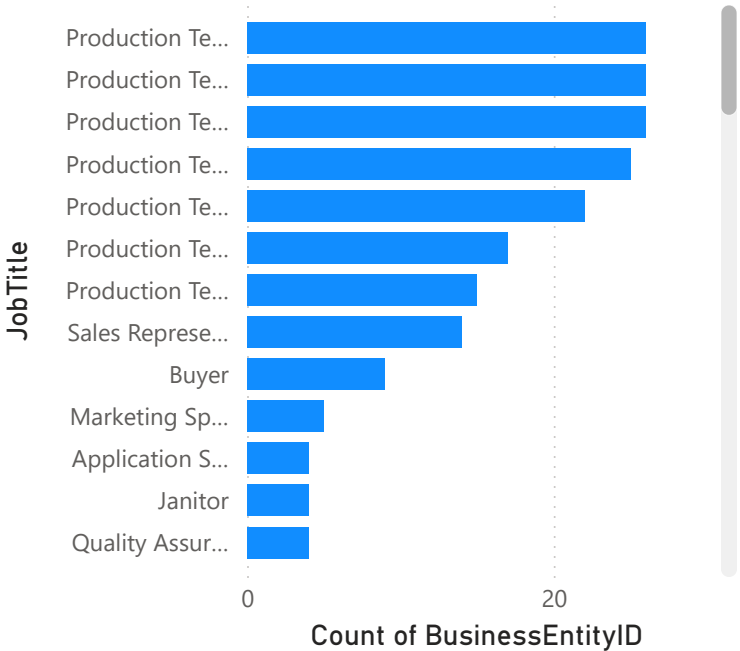
Count of BusinessEntityID by Gender



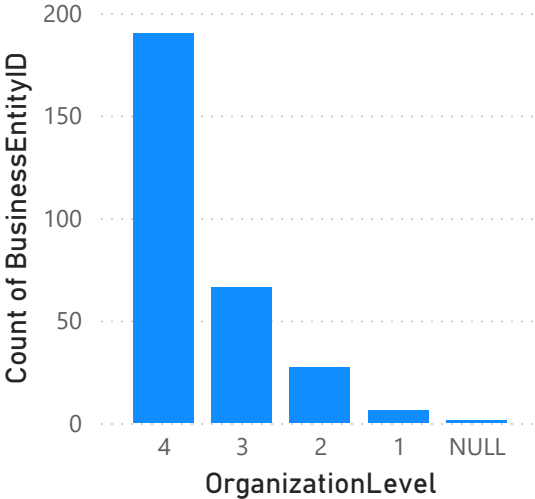
Count of BusinessEntityID by MaritalStatus

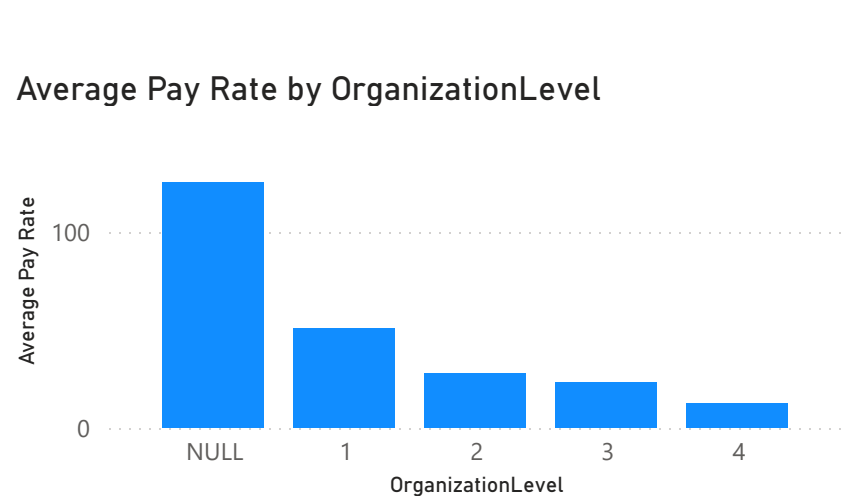
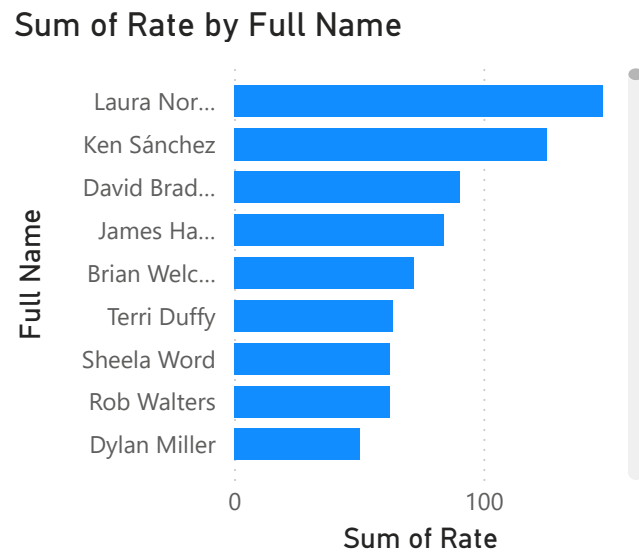
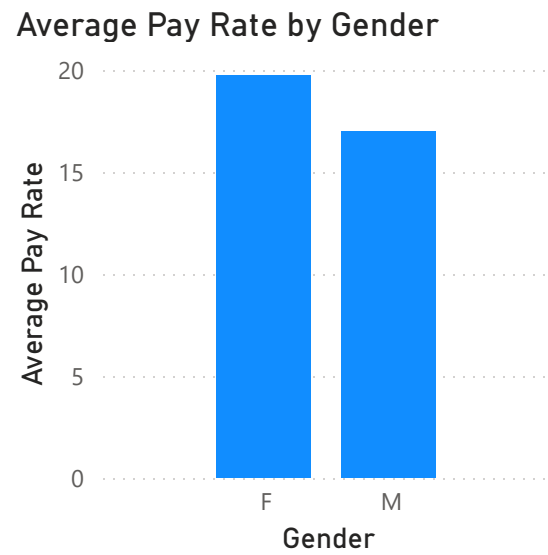
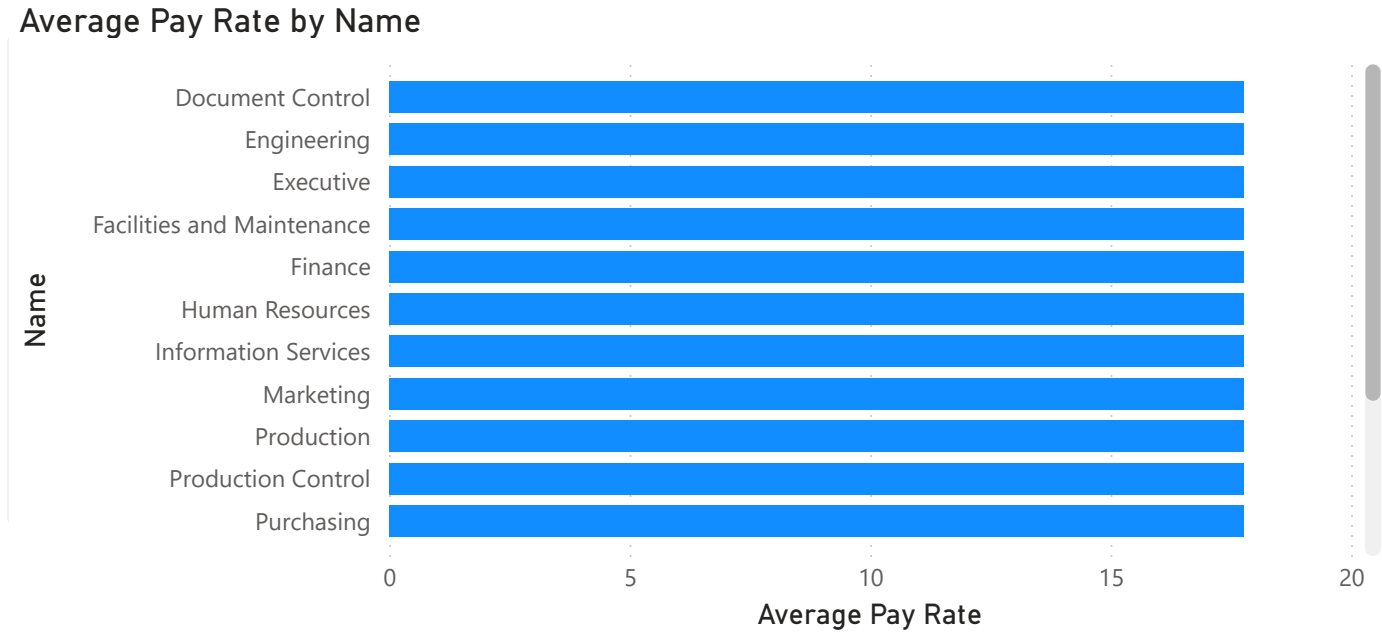
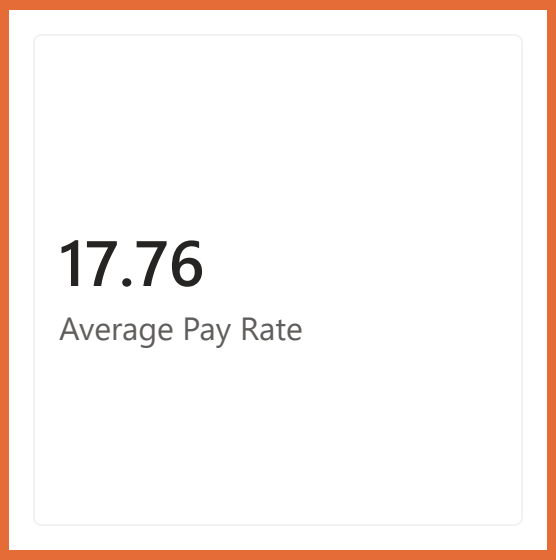
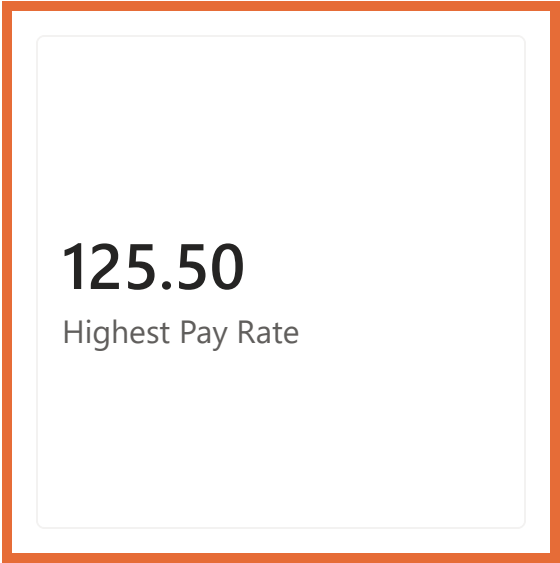


Employee by Job Tittle



Employee by Organization Level





17

Sales Employees

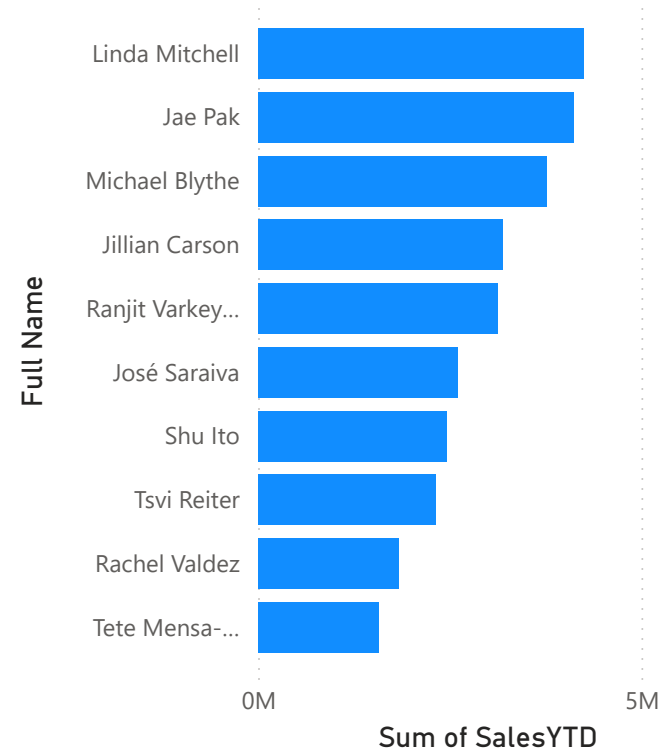
2.13M

Average Sales Per Employee

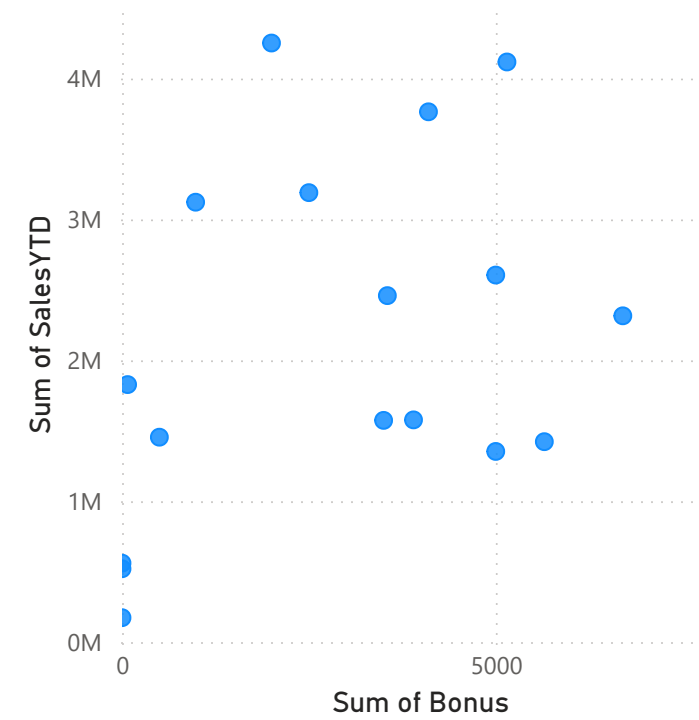
2.86K

Average Bonus

Sum of SalesYTD by Full Name



Sum of Bonus and Sum of SalesYTD by Full Name



50.61

Average Vacation Hours

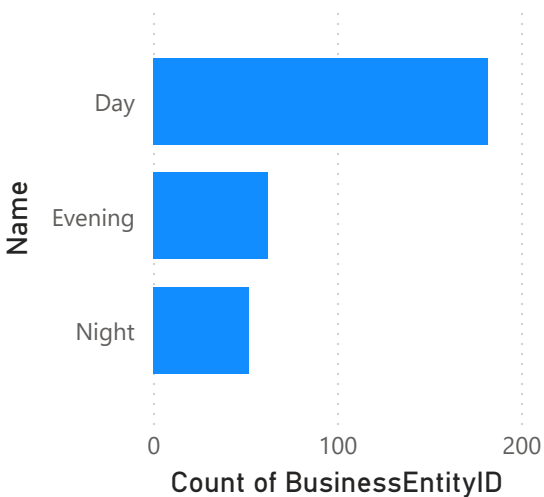
45.31

Average Sick Leave Hours

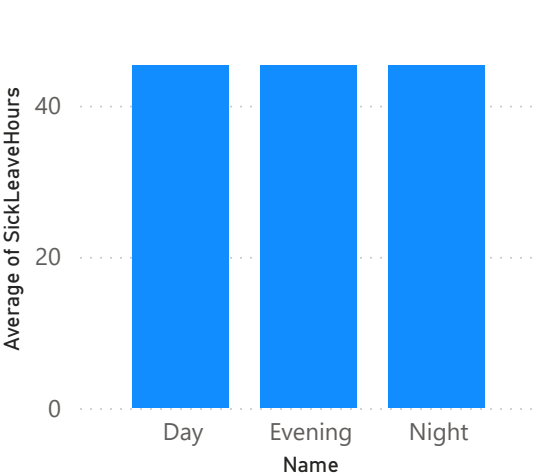
16.98

Average Years of Service

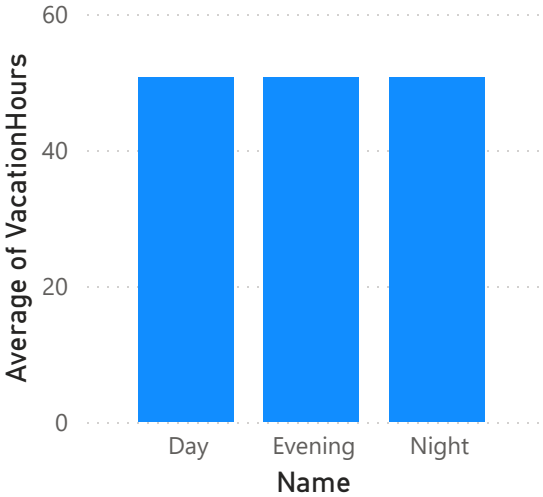
Count of BusinessEntityID by Name



Average of SickLeaveHours by Name



Average of VacationHours by Name



Count of Gender by Name and BusinessEntityID

